turbulence the radioactive "cloud" becomes quite diffuse. Therefore a more efficient technique called "total count" is used. A known amount (A_0) of the radionuclide is injected into the river and a downstream detector registers a total count (R_{tot}) as the radioactivity passes; the faster it passes, the lower is the measured radioactivity. Thus

$$F = \psi\, A_0 / R_{tot} \tag{18.38}$$

ψ is the counting efficiency which has to be determined under known conditions. The technique takes into consideration both longitudinal and transversal mixing.

18.6.3. Wear and corrosion

Wear and materials transfer are easily followed if the material undergoing wear is made radioactive. This has been used for studying wear of parts in automobile engines, cutting tools, ball-bearings, furnace linings, paint abrasions, etc. In this case it is important that the surface undergoing wear has a high specific radioactivity. If the material cannot be tagged by adding (e.g. plating) a radionuclide, the material has to be activated by irradiation (e.g. by accelerator) to produce the radioactive species.

Corrosion in gas and oil pipe-lines on the sea floor is monitored by welding patches containing a series of long-lived radionuclides at different material depth into the pipe wall during construction of the pipe-line. Wall corrosion will remove one isotope after the other as it proceeds. The presence of these isotopes can be monitored from time to time by sending a capsule with a suitable detector and recorder through the pipe.

18.6.4. Chemical processing

We have described some areas of application of radiotracers which are useful to the chemical industry. To be more general, radioactive tracers are used to seek information of flow and mixing patterns for

- parameter evaluations (mass balance, flow rates, kinetics)
- fluid dynamics (trouble shooting, residence time distributions, modelling flow phenomena)
- chemical reaction engineering (effect of flow and mixing on conversions, modelling of reaction systems, process control algorithms).

Each application is rather case specific and described in the chemical engineering literature.

18.7. ENVIRONMENTAL APPLICATIONS

Nuclear weapons tests have released large amounts of radionuclides into the atmosphere, which through their own weight or by rain have been carried to the earth's surface. Geophysics has made use of this weapon "fallout". By measurements on T (as HTO

water), ^{90}Sr, ^{137}Cs, and other fission products it has been possible to follow the movements of water from land via lakes and rivers into the sea, as well as to study the water streams of the oceans and the exchange between surface and deep water. As a result the circulation of water on earth has been mapped in quite detail. It has also been possible to analyze how tropical hurricanes are formed by measuring the water taken into the central part (the "eye") of the cyclone, since the HTO concentration in the normal atmosphere is different from that of surface ocean water due to isotopic effects.

Radioactive tracers like T_2O, $^{24}Na^+$, $^{82}Br^-$ ($\beta^-\gamma$ $t_{½}$ 1.471 d) and ^{51}Cr-EDTA (ECγ $t_{½}$ 27.704 d), are used in hydrology to determine the volume of natural water reserves (even underground) and to map the movement of ground and surface water as well as effluent pathways, Figure 18.17. Also the consumption of water and water flow in industries are readily determined, and leaking dams and pipes checked by radiotracers.

In order to avoid hazardous pollution it is important to discharge communal and industrial waste (silts, liquids, gases) so that the wastes are properly dispersed. The mapping of different dispersal sites is conveniently and commonly carried out by injecting a radioactive tracer at a testing spot, and then following its distribution at various depths, heights, and directions. In such a test it was found that sludge emptied into the River Thames at one point traveled upstream, which led to a repositioning of the sludge pipe exit. The simple technique of identifying hidden pipe leaks is illustrated in Figure 18.17 left. This technique is used not only for liquids, but also for checking the seal of underground electric cable hoses in which case gas tracers like ^{85}Kr ($\beta^-\gamma$ $t_{½}$ 10.72 y) or ^{133}Xe ($\beta^-\gamma$ $t_{½}$ 5.24 d) are used.

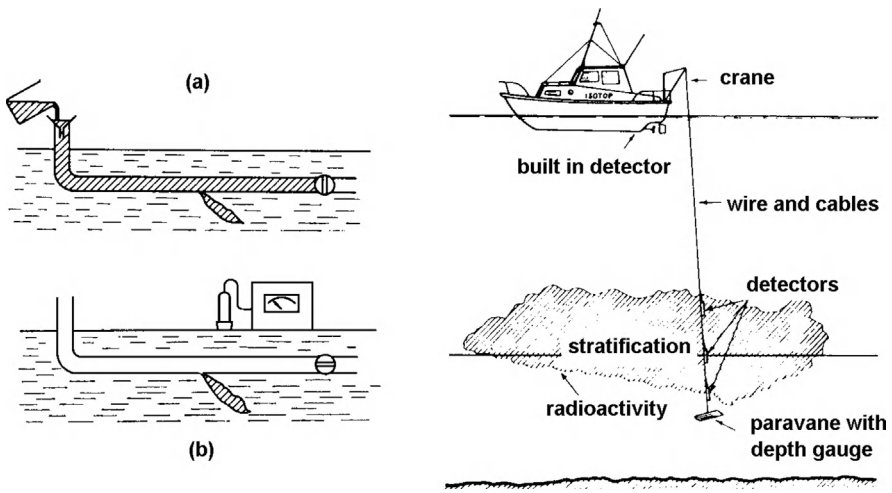


Figure 18.17 Left, technique for finding leaks in pipes. Right, technique for measuring stream lines in the strait of Öresund. *(Courtesy Danish Isotope Center.)*

Westermark and co-workers developed a unique method of revealing environmental history of natural rivers during the last 100–200 years. It is based on the freshwater pearl mussel *Margaritana margaritifera* living in these river. The mussel adds a new outer layer to its shell each year, incorporating into it trace metals (and also to some extent organics) dissolved in the water. The shell is cut and sectioned into small pieces, which are analyzed by α-track autoradiography, INAA and μ-PIXE. The latter is a "microprobe"-PIXE, i.e. the proton beam is focused by lenses to a resolution of a few microns. To get sufficient information the shell sample usually covers 3–15 years. With these techniques concentrations of some 30 elements have been determined retrospectively for >100 years in a number of Swedish rivers. Several unexplained trends have been observed, e.g. decreases in Ag, Au, Fe and Co. This is believed to be related to increasing air concentrations of SO_2. The observation of an increase in Br beginning in the 1940's is believed to be due to Br-additions to gasoline fuel. The technique has great potentials for increasing our knowledge on environmental changes of trace elements in nature. It could also be used as a long term biochemical control system at environmentally strategic places.

18.8. EXERCISES

18.1. The blood volume of a patient is to be determined by means of ^{32}P. For this purpose 15.0 ml of blood is withdrawn from the patient and mixed with a very small volume of $Na_2H^{32}PO_4$ of high specific activity. In 1 h the erythrocytes (red blood cells) take up all the ^{32}P; 1 ml is found to have an activity of 216 000 cpm in the detector system used. Exactly 5.00 ml of this tagged sample is reinjected into the patient, and 30 min later a new sample is withdrawn; 10 ml of this gives 2300 cpm. Calculate the blood volume.

18.2. A mixture of amino acids is to be assayed for cysteine. A 1.0 ml sample (density 1 g ml^{-1}) is withdrawn, and 2.61 mg of ^{35}S labelled cysteine of specific activity 0.862 μCi mg^{-1} is added. From this mixture pure cysteine is isolated by liquid-partition chromatography; 30.6 mg is isolated and measured to give 169 000 cpm in 27% detection efficiency. What is the percentage of cysteine in the original mixture?

18.3. In order to determine the lead content of a color pigment, 8.9871 g was dissolved in conc. HNO_3, and 5.00 ml ^{210}Pb solution added. After excess acid had been removed through evaporation, excess 1 M NaCl was added, the solution heated and filtered. After cooling and crystallization, the $PbCl_2$ was washed and recrystallized, 0.3276 g of the crystals was measured in a scintillation counter, giving 185 160 counts in 5 min. 1.000 ml of the original ^{210}Pb solution gave 57 000 cpm. The background was 362 cpm. Calculate the lead content (%) of the pigment.